

| STAT238 Lecture 1 Introduction to Bayesian Statistics

| 1 Introduction to Bayesian Statistics

| 1.1 Bayesian Statistics 是什么

根据 "Probability Theory – the Logic of Science" 书中的表述,

$$\text{Bayesian Statistics} = \text{Probability Theory}$$

| 1.2 Bayesian Statistics 如何运作

核心直觉: *Probability as Belief*

- 当我们无法完全确定一个命题是否为真时, 可以用一个 0 到 1 之间的概率来表示我们对其真实性的信念强度
- 概率的作用: 在不完全信息下, 对 "命题为真" 的可信程度进行定量表达

概率推断的两个基本步骤:

Step 1: Modeling Step

将已有/假设信息信息转化为 numerical assignments for the probabilities

Step 2: Inference Step

给定模型后, 使用 rule of probability 计算我们关心的事件的概率

Bayesian 语境中的表述:

- 未知真值通常表示为变量: $\theta \in \Theta$ for some set
- Inference step 涉及到计算 posterior probability:

$$P(\theta \in \Theta \mid \text{observed data and other background information})$$

- Modeling step 涉及到明确
 - likelihood:

$$P(\text{observed data} \mid \theta = \theta_0 \text{ and background information})$$

- prior model / prior distribution:

$$P(\theta = \theta_0 \mid \text{background information}) \quad \text{or} \quad f_{\theta \mid \text{background information}}(\theta_0)$$

| 2 Rules of Probability

🔗 Logic ▾

Probability 是分配给 propositions / events 的, 通常 probability 会 conditional on some information (包括已知的信息和假设的信息)

| 2.1 Conditional Probability

若 proposition A 的概率是基于某些 information I , 则记该条件概率为

$$(A | I)$$

⚠ Remark ▾

若 information I is clear from context, 则可以将条件概率 $(A | I)$ 简写为 (A) , 但我们仍需明确记住这些概率是基于某些信息的

| 2.2 Rule of Probability

1. **Basic Bounds:** $0 \leq (A | I) \leq 1$
2. **Product Rule:** $(A \ B | I) = (A | I)(B | A, I) = (B | I)(A | B, I)$
3. **Sum Rule:** 若 proposition A 可以被划分为 disjoint propositions $A_1, A_2, \dots$, 则 $(A | I) = \sum_i (A_i | I)$

基于以上 rules, 可以推出 Bayes rule:

$$\begin{aligned}(B | A, I) &= \frac{(A | B, I)(B | I)}{(A | I)} \\ &= \frac{(A | B, I)(B | I)}{(A | B, I)(B | I) + (A | B^c, I)(B^c | I)}\end{aligned}$$